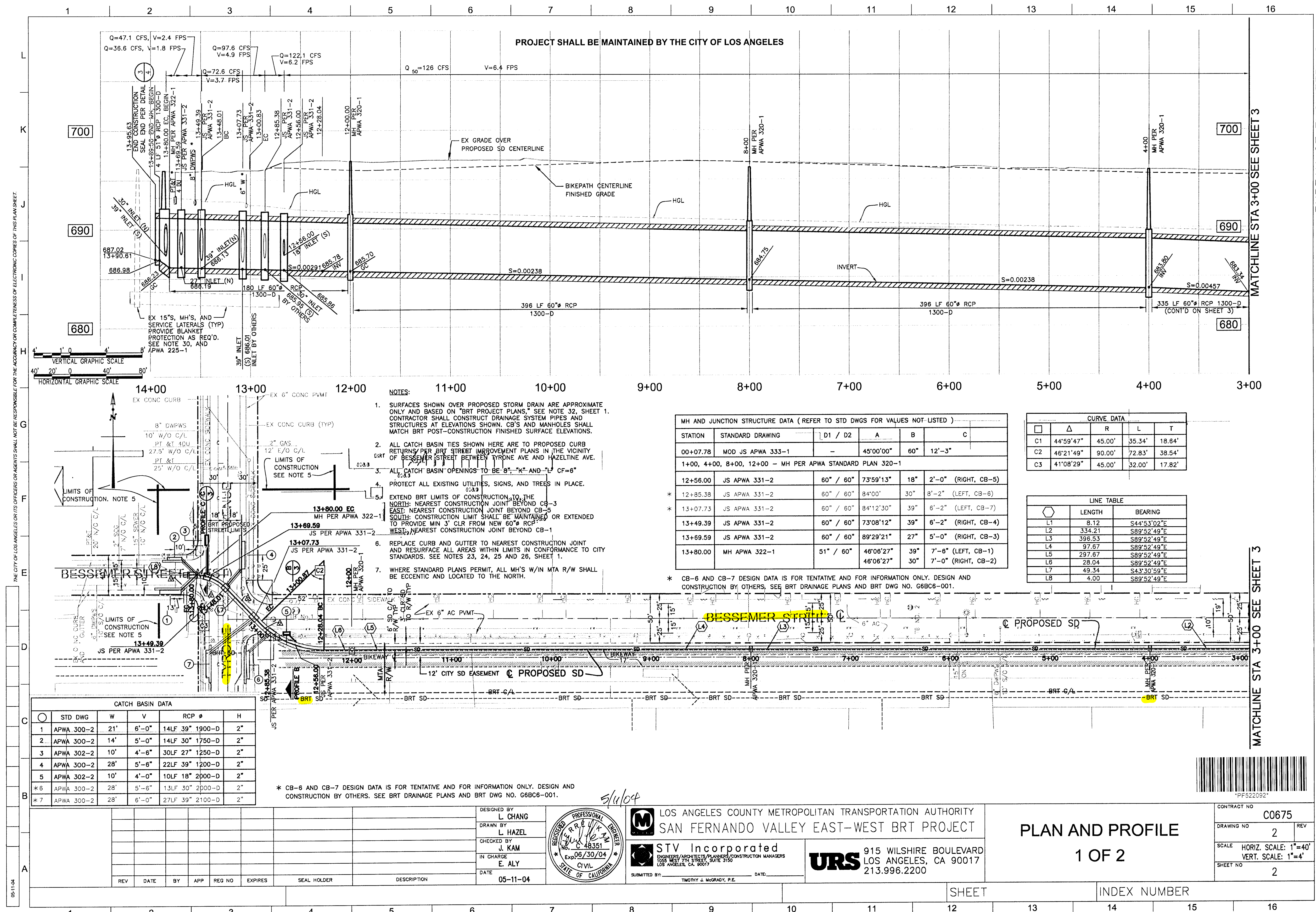




CATCH BASIN DATA					
STD DWG	W	V	RCP Ø	H	
1 APWA 300-2	21"	6'-0"	14LF 39" 1900-D	2"	
2 APWA 300-2	14"	5'-0"	14LF 30" 1750-D	2"	
3 APWA 302-2	10"	4'-6"	30LF 27" 1250-D	2"	
4 APWA 300-2	28"	5'-6"	22LF 39" 1200-D	2"	
5 APWA 302-2	10"	4'-0"	10LF 18" 2000-D	2"	
* 6 APWA 300-2	28"	5'-6"	13LF 30" 2000-D	2"	
* 7 APWA 300-2	28"	6'-0"	27LF 39" 2100-D	2"	

* CB-6 AND CB-7 DESIGN DATA IS FOR TENTATIVE AND FOR INFORMATION ONLY. DESIGN AND CONSTRUCTION BY OTHERS. SEE BRT DRAINAGE PLANS AND BRT DWG NO. G6B06-001.

- NOTES:
- SURFACES SHOWN OVER PROPOSED STORM DRAIN ARE APPROXIMATE ONLY AND BASED ON "BRT PROJECT PLANS." SEE NOTE 32, SHEET 1. CONTRACTOR SHALL CONSTRUCT DRAINAGE SYSTEM PIPES AND STRUCTURES AT ELEVATIONS SHOWN. CB'S AND MANHOLES SHALL MATCH BRT POST-CONSTRUCTION FINISHED SURFACE ELEVATIONS.
 - ALL CATCH BASIN TIES SHOWN HERE ARE TO PROPOSED CURB RETURNS PER BRT STREET IMPROVEMENT PLANS IN THE VICINITY OF BESSEMER STREET BETWEEN TYRONE AVE AND HAZELTINE AVE.
 - ALL CATCH BASIN OPENINGS TO BE 8", "K" AND "L" CF=6"
 - PROTECT ALL EXISTING UTILITIES, SIGNS, AND TREES IN PLACE.
 - EXTEND BRT LIMITS OF CONSTRUCTION TO THE NORTH: NEAREST CONSTRUCTION JOINT BEYOND CB-3 EAST; NEAREST CONSTRUCTION JOINT BEYOND CB-4 SOUTH; CONSTRUCTION LIMIT SHALL BE MAINTAINED OR EXTENDED TO PROVIDE MIN 3' CLR FROM NEW 60" RCP WEST; NEAREST CONSTRUCTION JOINT BEYOND CB-1
 - REPLACE CURB AND GUTTER TO NEAREST CONSTRUCTION JOINT AND RESURFACE ALL AREAS WITHIN LIMITS IN CONFORMANCE TO CITY STANDARDS. SEE NOTES 23, 24, 25 AND 26, SHEET 1.
 - WHERE STANDARD PLANS PERMIT, ALL MH'S W/IN MTA R/W SHALL BE ECCENTRIC AND LOCATED TO THE NORTH.

MH AND JUNCTION STRUCTURE DATA (REFER TO STD DWGS FOR VALUES NOT LISTED)					
STATION	STANDARD DRAWING	D1 / D2	A	B	C
00+07.78	MOD JS APWA 333-1	-	45'00"00"	60"	12'-3"
1+00, 4+00, 8+00, 12+00	-	-	-	-	MH PER APWA STANDARD PLAN 320-1
12+56.00	JS APWA 331-2	60" / 60"	73'59"13"	18"	2'-0" (RIGHT, CB-5)
* 12+85.38	JS APWA 331-2	60" / 60"	84'00"	30"	8'-2" (LEFT, CB-6)
* 13+07.73	JS APWA 331-2	60" / 60"	84'12'30"	39"	6'-2" (LEFT, CB-7)
13+49.39	JS APWA 331-2	60" / 60"	73'08'12"	39"	6'-2" (RIGHT, CB-4)
13+69.59	JS APWA 331-2	60" / 60"	89'29'21"	27"	5'-0" (RIGHT, CB-3)
13+80.00	MH APWA 322-1	51" / 60"	46'06'27"	30"	7'-0" (RIGHT, CB-2)

* CB-6 AND CB-7 DESIGN DATA IS FOR TENTATIVE AND FOR INFORMATION ONLY. DESIGN AND CONSTRUCTION BY OTHERS. SEE BRT DRAINAGE PLANS AND BRT DWG NO. G6B06-001.

CURVE DATA				
Δ	R	L	T	
C1 44°59'47"	45.00'	35.34'	18.84'	
C2 46°21'49"	90.00'	72.83'	38.54'	
C3 41°08'29"	45.00'	32.00'	17.82'	

LINE TABLE		
LENGTH	BEARING	
L1 8.12	S44°53'02"E	
L2 334.21	S89°52'49"E	
L3 396.53	S89°52'49"E	
L4 97.67	S89°52'49"E	
L5 297.67	S89°52'49"E	
L6 28.04	S89°52'49"E	
L7 49.34	S43°30'59"E	
L8 4.00	S89°52'49"E	

LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT

STV Incorporated
ENGINEERS/ARCHITECTS/PLANNERS/CONSTRUCTION MANAGERS
1000 WEST 7TH STREET, SUITE 3100
LOS ANGELES, CA 90017

URS
915 WILSHIRE BOULEVARD
LOS ANGELES, CA 90017
213.996.2200

DESIGNED BY: L. CHANG
DRAWN BY: L. HAZEL
CHECKED BY: J. KAM
IN CHARGE: E. ALY
DATE: 05-11-04

5/10/04

PLAN AND PROFILE

1 OF 2

CONTRACT NO.	C0675
DRAWING NO.	2
SCALE	HORIZ. SCALE: 1"=40' VERT. SCALE: 1"=4'
SHEET NO.	2

LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT

PROJECT: SFV EAST-WEST PROJECT
BESSEMER ST AND TYRONE AVE SD

ADDRESS: LOS ANGELES, CA

DESIGNED BY: L. CHANG
DRAWN BY: L. HAZEL
CHECKED BY: J. KAM
IN CHARGE: E. ALY
DATE: 05-11-04

LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT

PROJECT: SFV EAST-WEST PROJECT
BESSEMER ST AND TYRONE AVE SD

ADDRESS: LOS ANGELES, CA

DESIGNED BY: L. CHANG
DRAWN BY: L. HAZEL
CHECKED BY: J. KAM
IN CHARGE: E. ALY
DATE: 05-11-04

05-11-04

A. MODIFIED JUNCTION STRUCTURE BREAKOUT LIMITS SHALL BE DETERMINED AS FOLLOWS:

1. **UPSTREAM LIMIT** - APPROXIMATELY 6" UPSTREAM OF THE INTERSECTION OF THE OUTSIDE OF THE SPUR WALL WITH THE OUTSIDE WALL FACE OF THE EXISTING RCB (SEE PLAN VIEW)
2. **DOWNSTREAM LIMIT** - APPROXIMATELY 15' DOWNSTREAM OF THE INTERSECTION OF THE SPUR WALL WITH OUTSIDE WALL FACE OF THE EXISTING RCB (SEE PLAN VIEW)
3. **TOP AND BOTTOM LIMITS** - SEE UPSTREAM LIMIT DESCRIPTION. THE OPENING SHALL BE RECTANGULAR, CUT NORMAL TO THE WALL SURFACE AND WITHOUT DAMAGE TO REINFORCING STEEL
4. EXIST RCB REINFORCEMENT SHALL BE CLEANED, CUT AT CENTER OF OPENING AND BENT TOP AND BOTTOM OF JUNCTION STRUCTURE.
5. PRIOR TO REMOVING ANY PORTION OF THE RCB WALL, REMOVE EARTH LOADS ON TOP OF THE EXISTING RCB WITHIN THE RECTANGULAR AREA BETWEEN THE BREAKOUT LIMITS AND THE CENTERLINE OF THE BOX AND THE WALL
- B. WHERE REINFORCEMENT IS REQUIRED TO EXTEND THROUGH THE NEW JOINT, CONCRETE SHALL BE REMOVED IN THE FOLLOWING SEQUENCE:
 1. A SAWCUT SHALL BE MADE ONE AND ONE-HALF INCHES DEEP MAXIMUM AT THE BREAKOUT LIMIT. CARE SHALL BE EXERCISED IN SAWING AT THE REMOVAL LIMITS SO AS TO NOT CUT OR DAMAGE THE EXISTING REINFORCING STEEL IN THE RCB WALL. THE EXISTING REINFORCING STEEL SHALL BE RETAINED AND EXTENDED INTO THE NEW CONSTRUCTION AS INDICATED ON THE PLANS.
 2. USING HAND HELD EQUIPMENT, THE CONCRETE SHALL BE CAREFULLY REMOVED FOR THE FULL DEPTH OF THE RCB WALL WITHIN THE BREAKOUT LIMIT INDICATED ABOVE. EXISTING BARS SHALL BE CUT IN CENTER OF OPENING AND BENT INTO TOP, BOTTOM AND SIDES OF THE JUNCTION STRUCTURE AS SHOWN IN THIS SHEET.
 3. EXISTING REINFORCEMENT SHALL BE CUT TO THE REQUIRED BAR EXTENSION.
 4. THE CONCRETE MAY BE REMOVED BY ANY SUITABLE METHOD UPON APPROVAL OF THE ENGINEER, WHO SHALL BE THE SOLE JUDGE OF THE USE OF ANY CONCRETE REMOVAL EQUIPMENT, EXPLOSIVES, PORTLAND CEMENT OR OTHER SIMILAR DEVICES, WHICH ARE LIKELY TO DAMAGE THE CONCRETE, TO BE LEFT IN PLACE, SHALL NOT BE USED.

REINFORCEMENT

1. DETAIL AND PLACE ALL REINFORCING STEEL IN ACCORDANCE WITH ACI301 AND MANUAL OF STANDARD PRACTICE FOR DETAILING REINFORCED CONCRETE STRUCTURES (ACI315), AND ACCORDANCE WITH AREMA MANUAL.
2. REINFORCING BARS: ASTM A-615 GRADE 60 REINFORCING BARS TO BE WELDED: ASTM A-708
3. BEND BARS COL. SPlice REINFORCING ONLY WHERE INDICATED ON THE DRAWINGS IN ACCORDANCE WITH ACI 318-95.

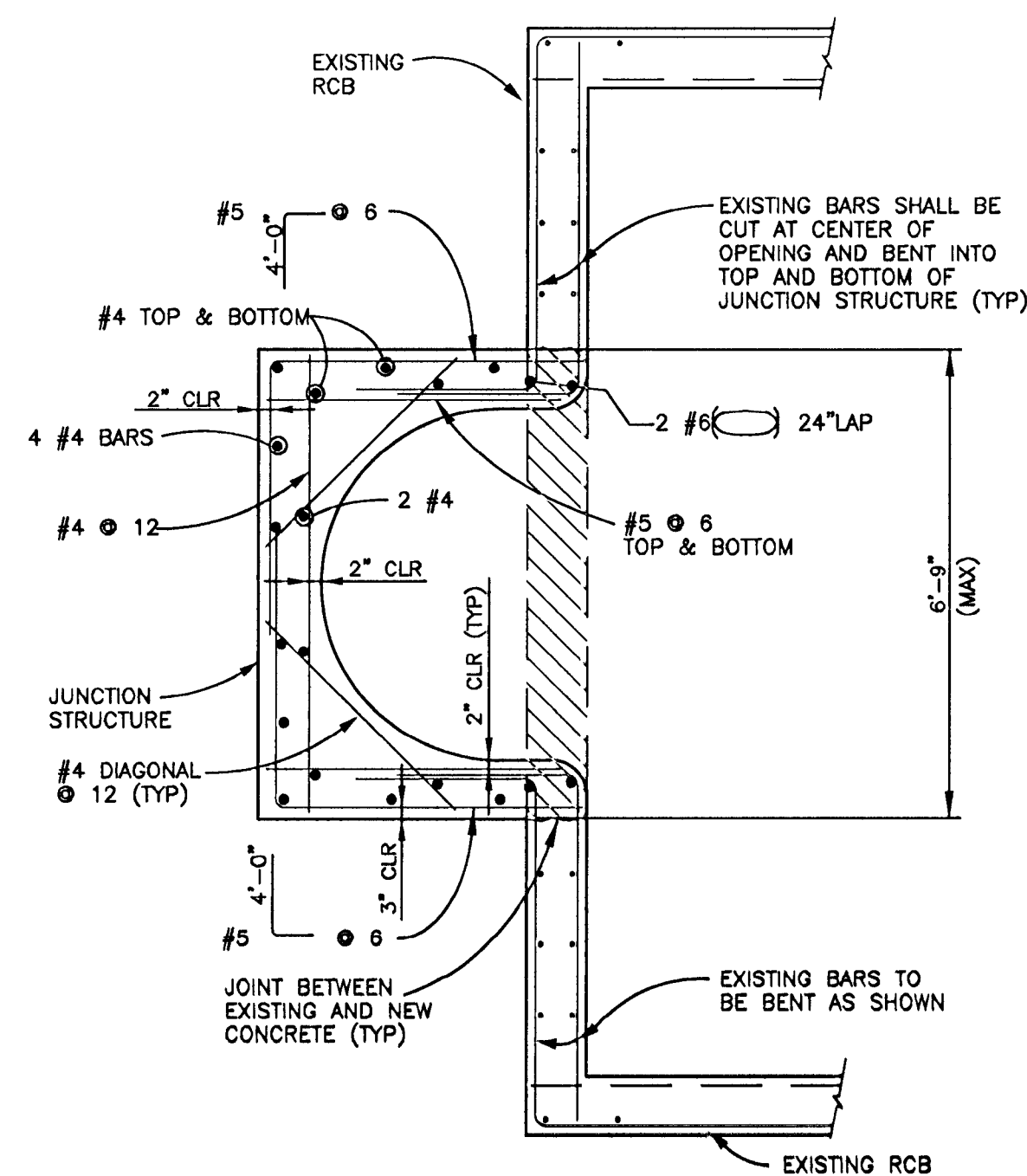
CONCRETE

1. ALL CONCRETE, CONCRETE WORK AND PLACEMENT OF REINFORCEMENT SHALL BE ACCORDANCE WITH CHAPTER 8 OF AREMA MANUAL SPECIFICATIONS.
2. THE ULTIMATE COMPRESSIVE STRENGTH OF CAST-IN-PLACE CONCRETE SHALL NOT BE LESS THAN 4000 PSI IN 28 DAYS. MAXIMUM SIZE OF AGGREGATE SHALL BE ONE INCH.
3. MINIMUM CONCRETE COVER TO THE BAR REINFORCEMENT SHALL BE TWO INCHES UNLESS NOTED OTHERWISE. THE COVER TO THE CONCRETE CAST AGAINST THE EARTH SHALL BE THREE INCHES.

DRILL AND BOND:

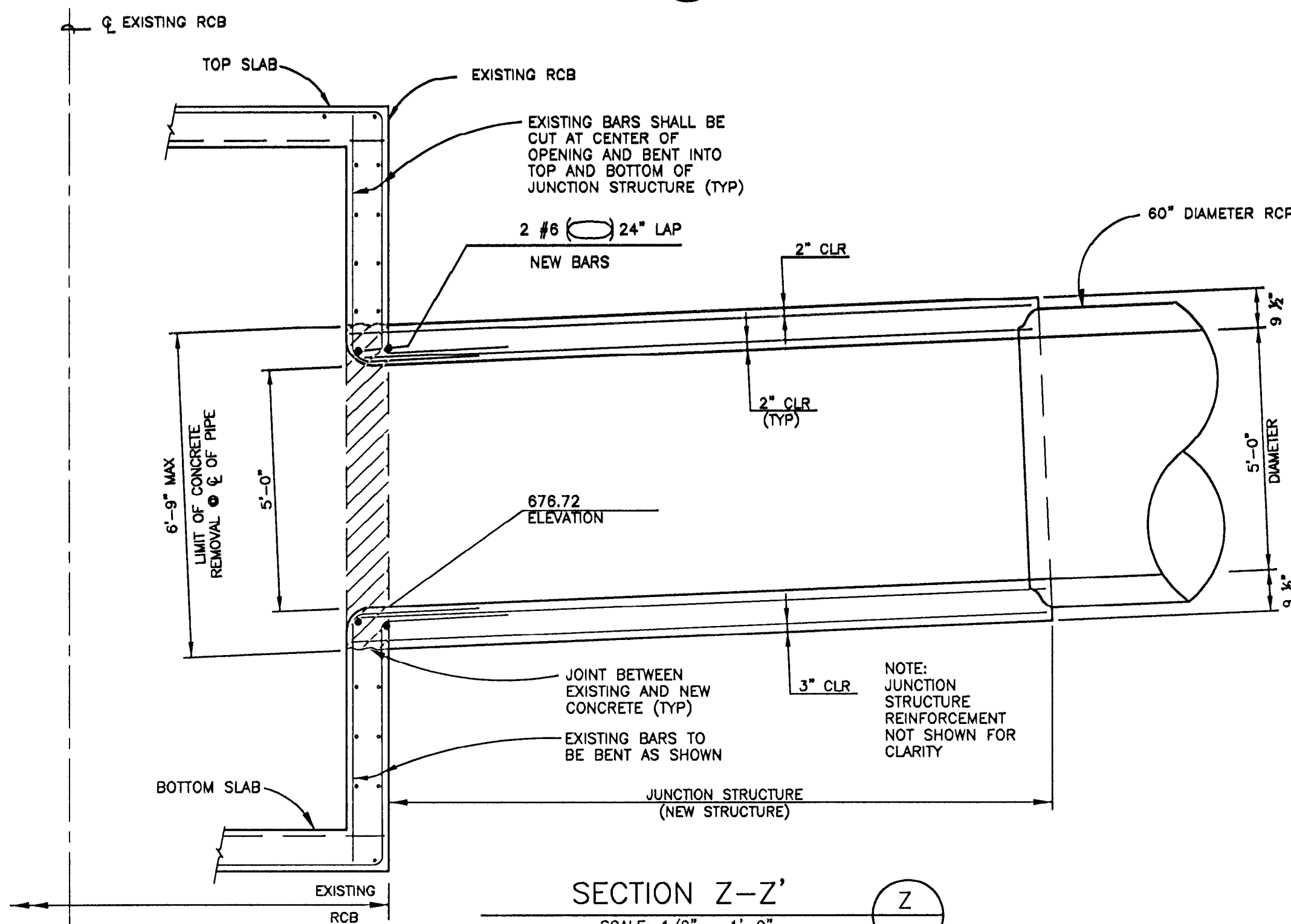
DRILL 5 HOLES AT BOTH SIDES OF WALL OPENING OF THE EXISTING RCB AT JUNCTION FOR #4 BARS WITH 6" EMBEDMENT @ 12". SLOPED 1:3 VERTICALLY, USE "DRILL AND BOND", AND EXTEND #4 BAR IN THE JUNCTION STRUCTURE AT LEAST 24". DRILL AND BOND PROCEDURE FOLLOWS L.A. COUNTY OR CALTRANS REQUIREMENTS.

FOR DETAILS NOT SHOWN ON THIS DRAWING,
REFER TO APWA STANDARD PLAN 333-1.



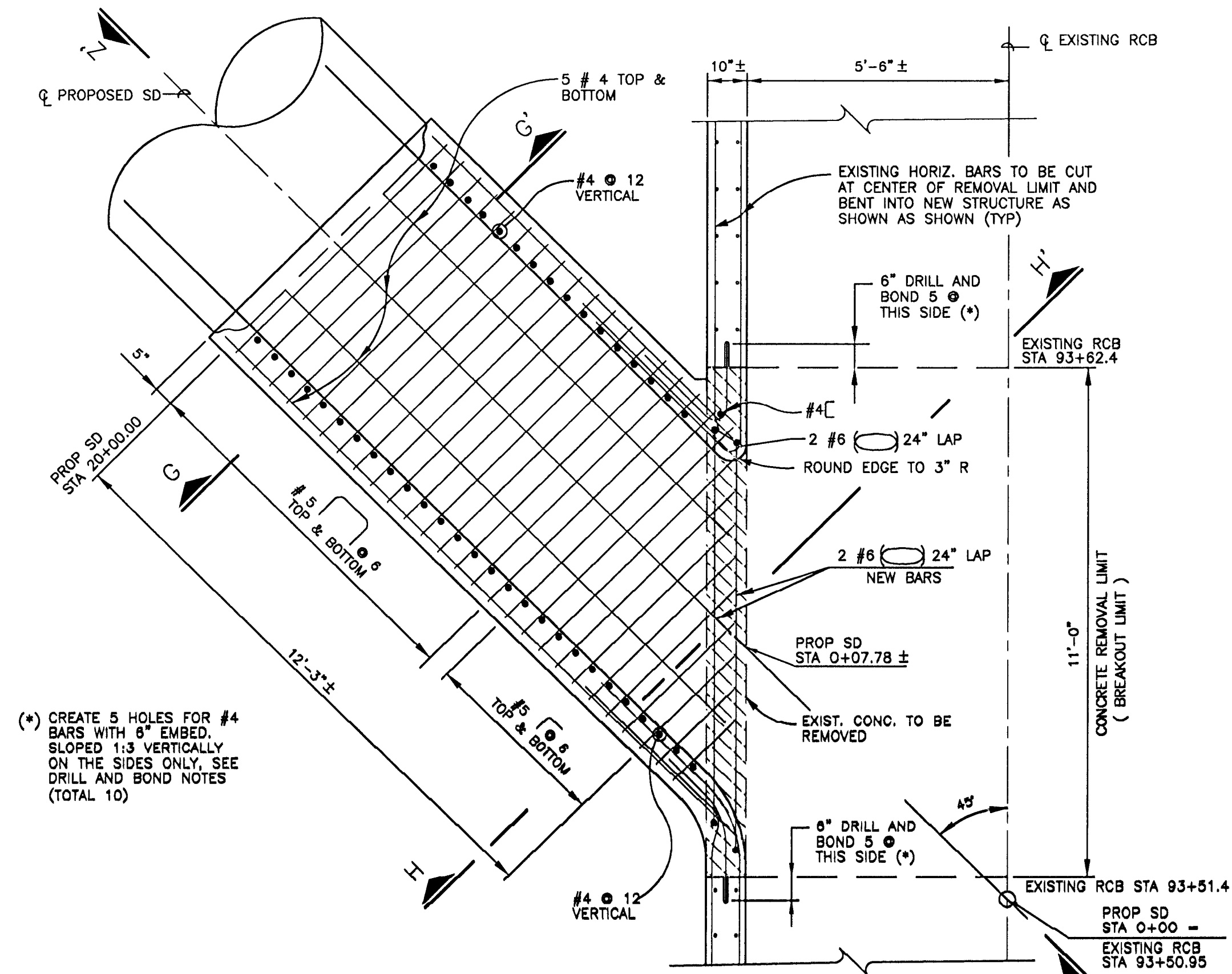
SECTION H-H

SCALE: 1/2" = 1'-0"



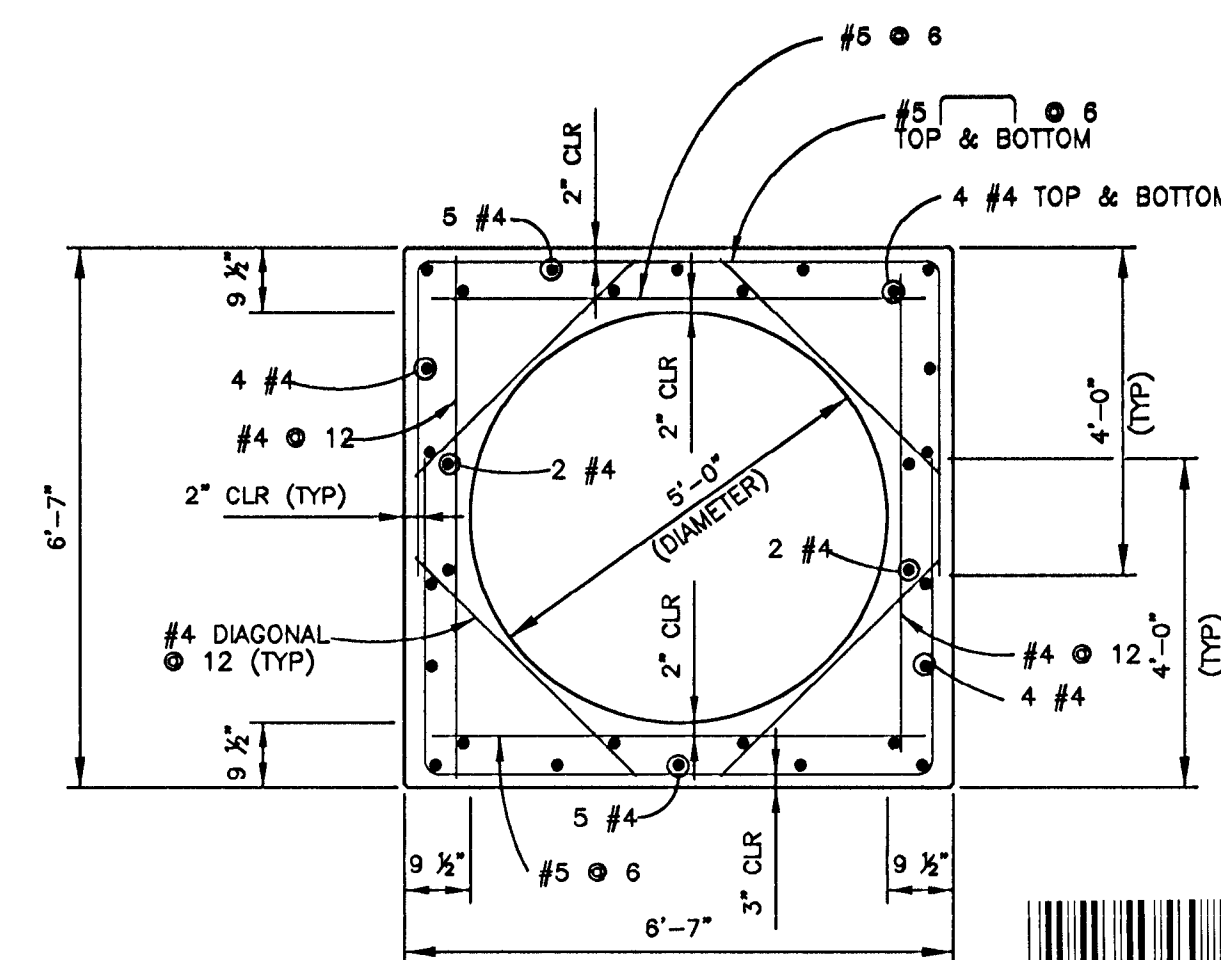
SECTION Z-Z'

SCALE: 1/2" = 1'-0"



PLAN VIEW

SCALE: 1/2" = 1'-0"

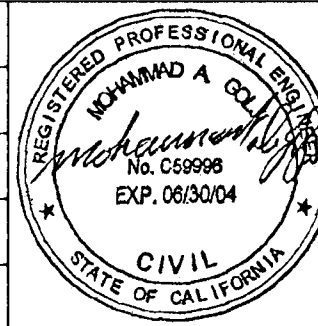


SECTION G-G

SCALE: 1/2" = 1'-0"

[illegible]

DESIGNED BY	M. GOLJI
DRAWN BY	L. GASTON
CHECKED BY	C. BAO
IN CHARGE	E. ALY
DATE	05-11-04



 LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT



STV Incorporated
ENGINEERS/ARCHITECTS/PLANNERS/CONSTRUCTION MANAGERS
1055 WEST 7TH STREET, SUITE 3150
LOS ANGELES, CA 90017

**PATTERSON
&
ASSOCIATES, INC.**
DESIGN-BUILD CONTRACTORS

725 TOWN & COUNTRY ROAD - SUITE 300
ORANGE, CA 92668

DETAILS

CONTRACT NO		C0675
DRAWING NO		4
SCALE		VARIES
SHEET NO		4

SHEET

INDEX NUMBER

BUREAU OF ENGINEERING

NO.	REVISIONS:	DATE:	BY:

	GARY LEE MOORE, P.E.	CITY
		ACCEPTED BY:

VERTICAL CONTROL:	NAVD88, 1988 ADJ
HORIZONTAL CONTROL:	NAD83, EPOCH 1995.0

LA DPW
ENGINEERING
WE PRESSURE
WAVING THE F

[illegible]

GARY LEE MOORE, P.E. CITY ENGINEER

ACCEPTED BY:

DATE

PLAN CHECKER

DATE

DIVISION / DISTRICT ENGINEER

DATE

VERTICAL CONTROL:	NAVDSB, 1989 ADJ
HORIZONTAL CONTROL:	NA0383, EPOCH 1995.0

SHEET TITLE:

DETAILS

PROJECT:

SFV EAST-WEST PROJECT
BESSEMER ST AND TYRON

ADDRESS:

LOS ANGELES, CA

PROJECT NO.	SZS11276
DRAWING NO.	

SHEET 4 OF 4 SHEETS